

PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
D1, D2, D3, D4, D6, D7, D8, D9 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R4, R6 - 1-mega-ohm
R5, R7 - 100-kilo-ohm

Capacitors:

- C1, C2 - 1000 μ F, 35V electrolytic
C3, C6 - 0.1 μ F ceramic disk
C4, C7 - 47 μ F, 25V electrolytic
C5, C8 - 0.01 μ F ceramic disk

Miscellaneous:

- RL1, RL2 - 12V, 1C/O relay
X1, X2 - 230V AC primary to 12V-0-12V/500mA secondary
CON1, CON2 - 5-pin connector interface for time switch
CON3, CON4 - 2-pin connector interface for starter switches
CON5 - 2-pin connector interface for AC input

Output of the monostable at pin 3 becomes high for a period equal to $1.1 \times R5 \times C4$, which is nearly equal to five seconds.

As pin 3 of IC1 is high, relay RL1 gets energised for five seconds, which, in turn, shorts the start switch, extending the 3-phase supply to the motor. This is virtually similar to physically pressing the start switch of the 3-phase motor starter for five seconds.

When time reaches 9am, the second time switch (stop switch) provides 230V AC to the primary of transformer X2. Again, by using a full-wave rectifier and filter circuit, 12V DC is provided to the second monostable circuit having relay RL2.

Normally-closed (N/C) terminal of the relay is connected in series with the stop switch of the starter of the 3-phase motor. So, the relay breaks the circuit to stop the motor.

This is an example of one time duration of 8am to 9am. In this



Fig. 3: Front of the time switch



Fig. 4: Rear of the time switch

way, a maximum of eight time durations can be programmed to switch the 3-phase electric motor on and off.

There is provision for setting days of the week for the controller to function. For instance, it can be set to work from Monday to Friday, Monday to Saturday, all seven days of the week, or only on a particular day of the week.

This system can find many applications, including switching on a water pump in a multi-storeyed commercial building to fill overhead tanks only for five or six days in a week. It can also prove useful for farmers, industrial units or railway stations where 3-phase motors are used.

Circuit and working

Two identical power supply circuits are built around transformers X1 and X2 with associated components as shown in Fig. 2. The arrangement provides 12V DC to two control circuits built around two 555 timers IC1 and IC2, which are configured in monostable mode.

The two time switches used in this system are Frontier-made, model TM-619-2. These operate on 230V AC at 50Hz. Each switch has a built-in single changeover relay with contact rating of 16A. It has an LCD display with buttons such as CLOCK, TIMER, DAY, HOUR, MIN and MANUAL, as shown in Fig. 3. By using these buttons, a real-time clock is set and various time durations are programmed.

The time switch is a programmable digital device that has a built-in real-time clock and can program for a maximum eight time durations. The time durations can be for a particular day, alternate days, Monday to Friday, Monday to Saturday, or Monday to Sunday.

By holding CLOCK button, real time is set by using HOUR, MIN and DAY buttons, while various time durations are programmed by using TIMER, HOUR, MIN and DAY buttons.

There are three modes, namely, ON, AUTO and OFF, written just below the display. After programming time durations, a black horizontal line segment is kept over AUTO mode from OFF mode by pressing MANUAL button. The time switch provides five external pins numbered 1 to 5 as shown in Fig. 4.

230V AC is applied across pins 1 and 2 of connectors CON1 and CON2 for start and stop switches, with

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